

EXTERNAL REVIEW DRAFT

Clean AI Receipts

Operational governance for secure knowledge models

How a machine readable receipt can turn AI governance from retrospective policy claims into inspectable evidence of authority, purpose, and provenance.



Review status

This document is unblessed until TCIEG review is complete. It describes a prototype demonstration and proposed governance architecture. It does not claim certification, legal compliance, or production deployment.

At a glance

THE DEMONSTRATION

A machine readable receipt for a defined AI lifecycle activity.

THE SECURITY EFFECT

Purpose bound access and a provenance reference for governed knowledge.

THE GOVERNANCE CHANGE

Evidence generated by operation, not only policy written afterwards.

THE PUBLIC PRINCIPLE

Verify who is accountable before the governed operation proceeds.

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Version notes

Personal contexts and personal information have been scrubbed from this report.

Executive summary

Artificial intelligence governance currently depends heavily on policies, assessments, model cards, and audit reports written about a system. These materials can be valuable, but they usually remain with the organisation making the claim and are reviewed after data has been collected, knowledge has been reused, or a model has been deployed.

The Clean AI receipt demonstration at itcop.org/paper/ tests a different approach. It represents governance as a machine readable receipt generated for a defined activity in the AI lifecycle. The receipt identifies the accountable participant, the scope of work, the applicable notice version, the authority relied on for data or knowledge reuse, relevant jurisdictions, upstream notice references, the current authorization state, and the integrity evidence needed to detect silent changes.

The demonstration does not expose model weights, dataset contents, trade secrets, or the identity of a person inspecting the record. It exposes the governance conditions surrounding the knowledge model.

This is the basis of **Clean AI**: an AI system that can produce independently inspectable evidence showing who is accountable, what knowledge was used, for which purpose, under which authority, and whether that authority remains current.

The significance for AI governance is practical. Governance can move from retrospective assertion to operational evidence, from one provider's final disclosure to a chain of accountable participants, and from broad claims about responsible AI to verifiable controls over knowledge contribution, access, reuse, and withdrawal.

1. Purpose

This report explains:

- how the AI receipt prototype demonstrates Clean AI;
- how receipts can support a secure knowledge model;
- how the approach changes the timing and evidence model of AI governance;
- what the prototype does and does not establish; and
- which questions external reviewers should test before further implementation.

2. The Clean AI proposition

Dirty AI asks for trust.

Clean AI carries evidence.

Clean AI is not a claim that a model is inherently safe, ethical, lawful, or accurate. It is an operating condition under which the authority, purpose, provenance, and accountable participants associated with the use of data and knowledge can be inspected and tested.

The core record is a Notice Receipt. For an AI lifecycle activity, the receipt can record:

- the receipt structure and version;

- the participant's role, such as designer, developer, trainer, or deployer;
- the activity and scope for which notice was given;
- the resolvable Controller Identification Record for the accountable participant;
- the immutable notice version applicable when the activity occurred;
- the lawful basis for personal data or the licence and terms governing published work;
- the upstream notices the participant received and relied on;
- the destinations and authority for relevant transfers;
- the current authorization state;
- an evidence request endpoint; and
- integrity evidence binding the receipt to the notice.

WHAT AN AI RECEIPT MAKES INSPECTABLE

01

Accountability

Participant role
Controller record
Scope of work

02

Authority

Purpose and lawful basis
Licence and terms
Current state

03

Evidence

Notice version
Upstream references
Integrity binding

A receipt containing these elements can answer three foundational questions about an AI system:

1. On whose authority was this material used?
2. Which notice and terms governed the activity at that time?
3. Where can the supporting evidence be inspected without requiring identification merely to ask?

3. The demonstration

The prototype should be understood as a compact demonstration of a record pattern, not as a complete AI governance platform.

It demonstrates that an AI governance record can be:

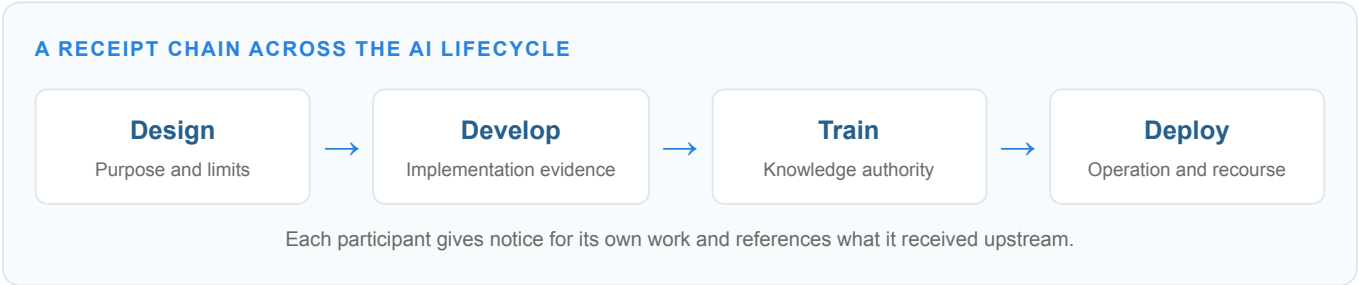
- **machine readable**, so systems and agents can evaluate it;
- **version bound**, so a later policy edit cannot silently change the historical claim;
- **participant specific**, so each contributor accounts for their own scope of work;
- **linked**, so downstream activity can resolve the notices received from upstream participants;
- **independently held**, so evidence does not remain solely with the organisation benefiting from it;
- **anonymous by default**, where inspection does not require a personal identifier; and
- **integrity protected**, so the notice and receipt cannot drift apart without detection.

The demonstration matters because a governance concept that cannot produce a record remains difficult to test. The prototype makes the proposed evidence object visible and gives reviewers something concrete to evaluate.

4. Chain of accountable participation

Modern AI systems rarely have one responsible actor. A system may involve a designer, developer, dataset provider, trainer, model provider, cloud operator, deployer, and downstream agent.

Clean AI assigns notice and evidence responsibilities across this chain.



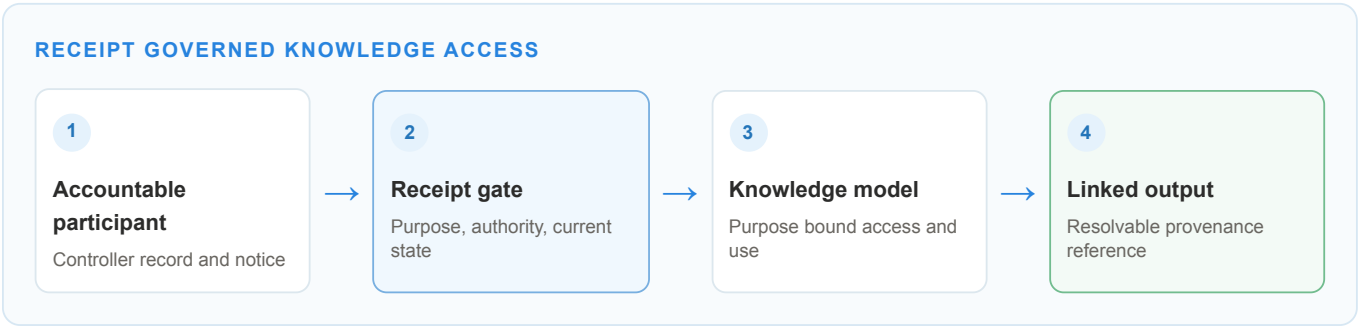
Participant	Scope recorded	Control enabled
Designer	Intended purpose, context, and known limitations	Defines the originating scope against which implementation can be tested
Developer	What was built and the design notice received	Records whether implementation remained within the stated design
Trainer	Corpus provenance, training purpose, lawful basis, licence, and reuse terms	Tests whether the authority relied on covers the material assembled
Deployer	Operational purpose, decision scope, oversight, and recourse	Tests whether the deployed system behaves within the notices received

Each participant’s receipt can reference the notices received from previous participants. This creates an inspectable chain of responsibility. If implementation departs from the declared design, if training exceeds the stated authority, or if deployment changes the purpose, the departure can be recorded and attributed where it occurs.

5. Enabling a secure knowledge model

A secure knowledge model needs more than technical protection around model files. It also needs governance security around the knowledge used to create and operate the model.

The receipt can support this through two connected functions.



5.1 Receipt as access condition

An AI agent or service presents a receipt bound to its accountable controller before accessing governed knowledge. The receipt is not a transferable password. It is bound to the registered participant and the approved scope.

Access decisions can therefore evaluate:

- who is requesting access;
- which purpose is declared;
- which knowledge class is requested;
- which assurance requirements apply;
- whether the governing notice and authority are current; and
- whether the requested operation remains inside the permitted scope.

Higher assurance material can require stronger evidence. Open material may remain available without a gate, while still carrying a provenance reference when reused.

5.2 Receipt as provenance seed

When governed knowledge is accessed, the receipt becomes the starting reference for later provenance. An output or banked knowledge item can carry a signed linking identifier that resolves to:

1. the governing receipt;
2. the accountable Controller Identification Record;
3. the applicable notice version;
4. the Notice Event Log; and
5. the current authorization state.

This linking reference can complement existing content provenance systems. It does not need to replace content credentials or watermarking. Its purpose is to connect those signals to the authority, purpose, and notice evidence governing the knowledge use.

5.3 The transparency invariant

The essential design rule is:

DESIGN PRINCIPLE

Gate use of governed knowledge, never inspection of the transparency records.

Controller Identification Records, notices, receipts, and relevant event records must remain inspectable. The governed material may be restricted according to purpose, authority, and assurance requirements, but accountability must not be hidden behind the same gate.

5.4 Security outcome

This produces a knowledge model with evidence concerning:

- origin and provenance;
- accountable participants;

- defined purposes;
- access conditions;
- lawful authority or licence;
- transfer context;
- traceable reuse;
- current authorization state;
- withdrawal or material change; and
- departures from declared scope.

Traditional cybersecurity protects systems and information from unauthorized access. Clean AI adds evidence showing whether the knowledge was authorized for the purpose in the first place.

6. How this changes AI governance

THE GOVERNANCE SHIFT	
CONVENTIONAL PATTERN	CLEAN AI PATTERN
Policy written about the system	Receipt generated by the operation
Evidence held by the provider	Evidence independently inspectable
Review after processing or harm	Evaluation before governed action
Responsibility concentrated at the edge	Accountability across participants

6.1 From policy claims to evidence of control

A policy describes an intended practice. A receipt records the governance claim attached to a specific activity, participant, purpose, and version. This gives another party an evidence object that can be checked.

6.2 From retrospective review to operational evaluation

Current oversight often begins after collection, training, deployment, or harm. A machine readable receipt can be evaluated before a system trains, retrieves knowledge, deploys a model, or executes an agentic action.

Where the accountable participant, purpose, authority, or current authorization state cannot be resolved, the operation can be stopped or referred for review. Governance becomes part of execution rather than an investigation conducted afterwards.

6.3 From one final disclosure to chain accountability

The organisation facing the individual often provides the final disclosure for a system assembled by many parties. The receipt model requires each participant to account for the work actually performed. This reduces the risk that responsibility disappears inside a complex supply chain.

6.4 From content provenance to authority provenance

Content provenance can show where a file came from and how it changed. Clean AI also asks:

- Was the material authorized for this use?
- Which lawful basis, licence, or terms applied?
- Which purpose was declared?
- Is the authority still current?
- Who is accountable for the decision to use it?

This adds permission and authority provenance to content provenance.

6.5 From general models to purpose bound knowledge models

The more useful governance object is often the knowledge model: a body of knowledge assembled for a defined domain, jurisdiction, community, or public purpose.

A purpose bound model can make its access conditions and reuse authority inspectable. This is relevant to health, education, research, cultural knowledge, public administration, and other settings where provenance and authority are inseparable from the legitimacy of the resulting model.

6.6 From ethical branding to an inspectable market signal

Responsible AI claims are difficult to compare when every provider can use similar language. A producible receipt offers a more disciplined signal.

A Clean AI claim should mean that the provider can produce the applicable notice, authority, provenance, participant, and current state evidence. Markets, procuring bodies, regulators, insurers, and affected individuals can then evaluate evidence rather than branding.

7. Relationship to Digital Transparency Infrastructure

The receipt is not a separate AI protocol. It applies a standard record and receipt pattern to AI lifecycle activities.

The wider Digital Transparency Infrastructure consists of interoperable accountability records, including:

- a Controller Identification Record;
- a versioned Notice Record;
- a Notice Receipt;
- a Notice Event Log; and
- a current authorization state.

The infrastructure provides transparency. The outcome is privacy enabling control, accountable knowledge reuse, and regulator scale inspection.

For public communication, the sequence is simple:

Verify who is accountable, inspect the terms, receive the receipt, and only then permit the governed operation.

8. What the prototype does not establish

The demonstration should not be represented as proving that:

- the full receipt network is deployed;
- the prototype certifies an AI system as lawful, safe, accurate, or ethical;
- every receipt field has an agreed international vocabulary;
- a recorded lawful basis is necessarily valid;
- provenance evidence requires disclosure of datasets, model weights, or trade secrets;
- receipt possession alone authorizes unrestricted reuse; or
- the design has completed independent security, legal, and interoperability testing.

A receipt records the authority claimed and the participant making the claim. Its value is that the claim becomes testable and contestable at the point it is asserted.

9. External review questions

Reviewers are invited to test the following questions.

Governance

1. Does the receipt identify the minimum evidence needed to evaluate accountability, purpose, authority, and provenance?
2. Are responsibilities correctly divided across designers, developers, trainers, deployers, and agents?
3. Which claims require independent verification rather than self assertion?

Security

1. How should receipts be bound to controllers and agents to prevent transfer, replay, or impersonation?
2. How should revocation, suspension, and material change propagate through downstream knowledge use?
3. Can a provenance reference be removed, downgraded, or detached without detection?

Privacy and rights

1. Can inspection and evidence requests remain available without forced personal identification?
2. How should withdrawal, objection, licence changes, and purpose changes affect trained and deployed models?
3. What information must remain private while still allowing independent governance inspection?

Standards and interoperability

1. Which receipt elements can be expressed through ISO/IEC TS 27560 and its proposed extensions?
2. Which vocabularies are needed for participant roles, purposes, transfer mechanisms, licences, and authorization states?
3. How should the receipt link to content provenance credentials and existing AI management systems without creating a competing format?

Implementation

1. What is the smallest prototype that can test controller registration, receipt issuance, receipt bound access, state change, and provenance resolution?
 2. Which public interest knowledge environment would provide the strongest first use case?
 3. What evidence threshold should support a public Clean AI claim?
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10. Recommended next actions

1. Conduct external standards, security, privacy, and implementation review of the receipt fields and chain model.
 2. Publish a stable, versioned example receipt with a human readable explanation and machine readable representation.
 3. Define the minimum conformance test for controller resolution, notice version binding, authority evidence, and anonymous inspection.
 4. Threat model controller binding, replay, transfer, stripping, downgrade, revocation, and false provenance claims.
 5. Test a narrow knowledge access demonstration in which an agent registers, receives a receipt, accesses a governed resource, and produces an output with a resolvable provenance reference.
 6. Separate clearly the open transparency records from access controlled knowledge resources.
 7. Document the relationship to ISO/IEC TS 27560, ISO/IEC 29184, ISO/IEC 27091, ISO/IEC 42001, Convention 108+, the EU AI Act, and existing content provenance specifications.
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11. Source material

- [ITCoP AI receipt prototype and paper](#)
 - ISO/IEC 29184, *Online privacy notices and consent*
 - ISO/IEC TS 27560:2023, *Consent record information structure*
 - ISO/IEC 27091, *Privacy protection for AI systems*
 - ISO/IEC 42001:2023, *Artificial intelligence management system*
 - Convention 108+
 - Regulation (EU) 2024/1689, Artificial Intelligence Act
 - Coalition for Content Provenance and Authenticity, Content Credentials
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12. Version notes

v0.1 External Review, 1 September 2026

- Converted the Clean AI explanation into a TCIEG report structure.
- Applied public terminology and entity separation rules.
- Removed internal development language and unsupported production claims.
- Added explicit prototype boundaries and external review questions.
- Marked the report unblessed pending TCIEG review.